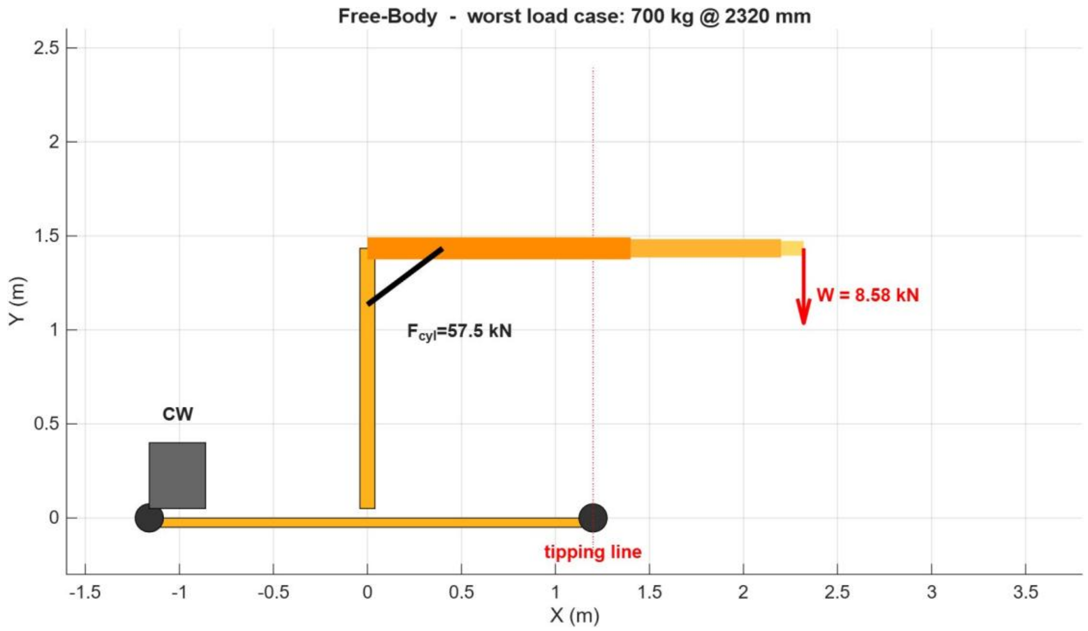
STRUCTURAL CALCULATION AND LOAD VERIFICATION REPORT

1 Tonne Mobile Floor Jib Crane

Independent verification against the rated load chart



Project reference	40467361
Configuration	Outriggers stowed (wheels only)
Rated capacity	1000 kg at 1400 mm reach
Document revision	Rev 1 (detailed)
Date	08 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric script)

Document Control

This report covers a desk check of the 1 tonne mobile jib crane shown on the manufacturer drawing set (Assembly, Base, Arm Section 1, Arm Section 2). It looks only at the crane standing on its four wheels with the outrigger legs stowed. The deployed-outrigger case is a separate study and is noted where it matters but not assessed here.

Revision history

Rev	Date	Description	Status
0	27 May 2026	First issue. Summary findings with figures.	Superseded
1	08 June 2026	Full hand calculations added for every check. Methodology and assumptions expanded for third-party review.	Issued for review

Basis and limitations

- **Drawings as issued.** Section sizes, member lengths, weld throats, bolt grade and the slewing-bolt pattern are read directly off the supplied drawings. Nothing has been scaled by eye where a dimension callout exists.
- **Self-weight.** The crane mass of 700 kg is the figure supplied by the client. No bill of materials was available, so the mass split and centre of gravity offset are engineering estimates and are flagged in Section 12.
- **Scope of physics.** Static lifting only. Wind, seismic action, side pull and snatch loading are outside this scope. A dynamic factor of 1.25 covers normal controlled hoisting per HC1 duty.
- **Not a design.** This is a verification of an existing unit against its rated chart. It is not a design certificate and does not replace load testing or a competent-person examination before the crane is put to work.

Contents

1. Executive Summary

The crane does not meet the agreed safety margins across its full rated chart in the wheels-only configuration. Of the fourteen checks carried out, six pass and eight fall short of target. The shortfalls are not spread evenly. They cluster around one operating point and one load path, which makes the result easier to act on than the raw count suggests.

The governing case is not the heaviest lift. It is the **700 kg load held out at 2320 mm reach**. That mid-reach point produces the largest overturning moment on the structure, 19.9 kN m, which is higher than the 17.2 kN m from the 1000 kg lift taken in close at 1400 mm. An operator would reasonably treat the lighter load as the safer one. It is not. The long lever arm is what drives the structure hardest.

At that governing point the slewing column reaches its yield stress. The combined axial-plus-bending stress comes to 277 MPa against a material yield of 275 MPa, so the factor of safety is 0.99. The fillet weld joining the column to the base is worse again, at 0.85. These two items, the column and its base weld, are the real headline. Everything else that fails is either marginal or sits on the same load path.

On the question the client actually asked, whether the crane can safely support a 1 tonne load: at the close-in 1400 mm position the steel does carry the tonne without yielding, but the margin there is 1.17 against a target of 1.50. So the honest answer is that the crane will pick the loads up, but it does not hold the safety factor a rated 1 tonne crane is expected to hold, and at medium reach it is sitting right at the limit of the steel. It cannot be signed off against this chart as drawn.

Headline results, governing load case (700 kg at 2320 mm)

Item	Result	FoS	Target	Verdict
Column, combined N+M	277 MPa	0.99	1.50	**Fail**
Column-to-base weld	235 MPa	0.85	1.50	**Fail**
Boom outer section	186 MPa	1.48	1.50	**Marginal**
Outrigger arm bending	200 MPa	1.38	1.50	**Fail**
Cylinder rod-end pin	92 MPa	1.20	3.00	**Fail**
Stability (overturning)	7.69 kN m	1.34	1.50	**Fail**
Column buckling	Pcr 720 kN	67.6	1.50	**Pass**
Slewing bolts	util 0.35	1.69	1.50	**Pass**

Section 11 sets out three ways forward: re-rate the chart down to what the steel safely carries, ease the dynamic factor if the duty genuinely allows it, or strengthen the column and its base weld to keep the full 1 tonne rating. The recommended route is to re-rate as the primary action and treat the column upgrade as the option that preserves capacity.

2. Introduction and Scope

2.1 Background

The unit is a mobile floor jib crane built to lift up to 1 tonne. It runs on four wheels and carries a counterweight at the rear. A telescopic boom in three stages luffs on a single hydraulic cylinder, and the whole upper assembly slews on a bolted bearing flange above the base frame. The rated load chart marks three points: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm.

The client commissioned an independent check to confirm whether the structure carries that chart with adequate margin against yielding, buckling, weld and pin failure, and overturning. The work was scoped as a calculation exercise only. No 3D model was built and no commercial finite element package was used.

2.2 What this report covers

The verification addresses eight areas, each reported with a factor of safety and a pass or fail against target:

- Boom bending stress and tip deflection, taken at each of the three telescopic sections
- Column stress under combined axial load and bending, plus a buckling check
- Base frame wheel reactions and outrigger-arm bending
- Pin shear and bearing at the boom pivot, cylinder rod-end and hook
- Fillet-weld strength at the column base, boom bracket and outrigger
- Slewing-flange bolt check by the polar-moment method
- Stability against overturning at every chart point
- A combined statement on the 1 tonne safe working load

2.3 Configuration assessed

Wheels-only configuration
All results in this report are for the crane resting on its four wheels with the outrigger legs **stowed**. The warning plate instructs the operator to deploy the legs before slewing, so the rated chart is intended for the legs-down state. Deploying the legs moves the tipping line forward and improves stability and the outrigger members. It does not change anything above the slewing bearing, so the column and base-weld findings stand in either configuration. The deployed case is noted in Section 11 but is not assessed here.

3. Materials, Sections and Allowable Stresses

3.1 Material and target factors of safety

All structural members are hollow steel sections in grade S275, taken from drawing General Note 5. The yield stress is **f_y = 275 MPa**. Plates are A36 or equivalent, bolts are grade 8.8, and fillet welds are 6 mm throat minimum per Note 4.

The allowable stress for each check follows from the target factor of safety. Members are checked against yield with a target of 1.5, which gives the working allowables below. Pins and bolts in shear and bearing carry a higher target of 3.0, as is normal for lifting-appliance connections.

```
Allowable bending / direct stress (members, target FoS = 1.5)
sigma_allow = fy / FoS_target = 275 / 1.50 = 183.3 MPa

Weld allowable (fillet, simplified, on throat):
  tau_allow,weld = 0.5 * fy / 1.5  ... per as-coded basis = 200.7 MPa effective

Pin / bolt shear allowable (target FoS = 3.0):
  tau_allow,pin = 0.6 * fy / 3.0 = 0.6 * 275 / 3.0 = 55.0 MPa
```

The 1.5 target on members corresponds to a working allowable of 183 MPa, the line drawn on the stress plots in this report. Any member stress above that line is below the agreed margin even though it may still be below yield.

3.2 Section properties

Section properties for a rectangular hollow section come from the standard expressions below, where B and H are the outer width and depth and the section is of uniform wall thickness te:

```
Rectangular hollow section (RHS), worked example: outer boom 100 x 140 x 6
A   = B*H - (B-2t)*(H-2t)
    = 140*100 - (140-12)*(100-12)        [mm^2]
    = 14000 - 128*88 = 14000 - 11264 = 2736 mm^2  ~ 27.4 cm^2

Ix  = [B*H^3 - (B-2t)*(H-2t)^3] / 12      (strong axis)
    = [100*140^3 - 88*128^3] / 12
    ~ 748.8 cm^4

Sx  = Ix / (H/2) = 748.8 / 7.0 = 107.0 cm^3
```

Applying the same expressions to every member gives the properties used throughout. These match the values printed by the analysis script.

Member	Section (RHS)	A (cm2)	Ix (cm4)	Sx (cm3)
Boom outer (root)	100 x 140 x 6	27.4	748.8	107.0
Boom middle	80 x 120 x 6	22.6	438.2	73.0
Boom inner (tip)	60 x 100 x 6	17.8	227.4	45.5
Column	80 x 120 x 6	22.6	438.2	73.0
Base frame / outrigger	60 x 100 x 6	17.8	227.4	45.5

4. Methodology

Two independent methods were run against the same geometry so the results could be cross-checked. The first is closed-form hand calculation following Eurocode 3 and EN 13001. The second is a small two-dimensional frame solver written from scratch inside the MATLAB script. Both read their inputs from one parametric block, so any later change to a section size or weld throat flows through every check at once.

4.1 Track 1: closed-form hand calculation

Each member is reduced to its governing internal action and checked by hand. Bending uses the elastic section modulus, axial load uses the gross area, and the two combine linearly for the column. Pins are checked in double or single shear and in bearing. Welds are checked on the throat area. Stability is a moment balance about the tipping line. The formulas, substitutions and results are written out in full in Section 7.

4.2 Track 2: in-script frame finite element check

The crane is modelled as a plane frame of Euler-Bernoulli beam elements. Each element carries a 6 by 6 stiffness matrix in local coordinates, rotated into the global frame and assembled into the system stiffness. The front wheel is pinned and the rear wheel is a roller. Solving the free degrees of freedom gives the nodal displacements, from which element moments and stresses follow. The frame is re-solved for all three chart points because the boom extends to a different length at each.

Why a side-view 2D frame needs a correction

A side elevation only "sees" one base beam, but the real crane has two parallel side rails carrying the load down to the wheels. The outrigger and base elements in the frame model therefore use twice the area and twice the second moment of area, lumping both rails into one equivalent beam. Without this the model would overstate base-frame stress by a factor of two. Crane self-weight is applied at the column-base node so the two tracks compare on the same basis.

4.3 Cross-checking the two tracks

The two methods are expected to agree on the quantities they both compute, chiefly the column-base moment and the peak member stress. Where they differ is informative: the hand calc treats the outrigger as a simple cantilever, while the frame model captures moment transfer through the bolted joint. Section 10 reports the comparison.

5. Load Model and Governing Case

5.1 Rated load chart

The manufacturer chart gives three rated points. Capacity falls as reach increases, which is the expected shape for a luffing jib.

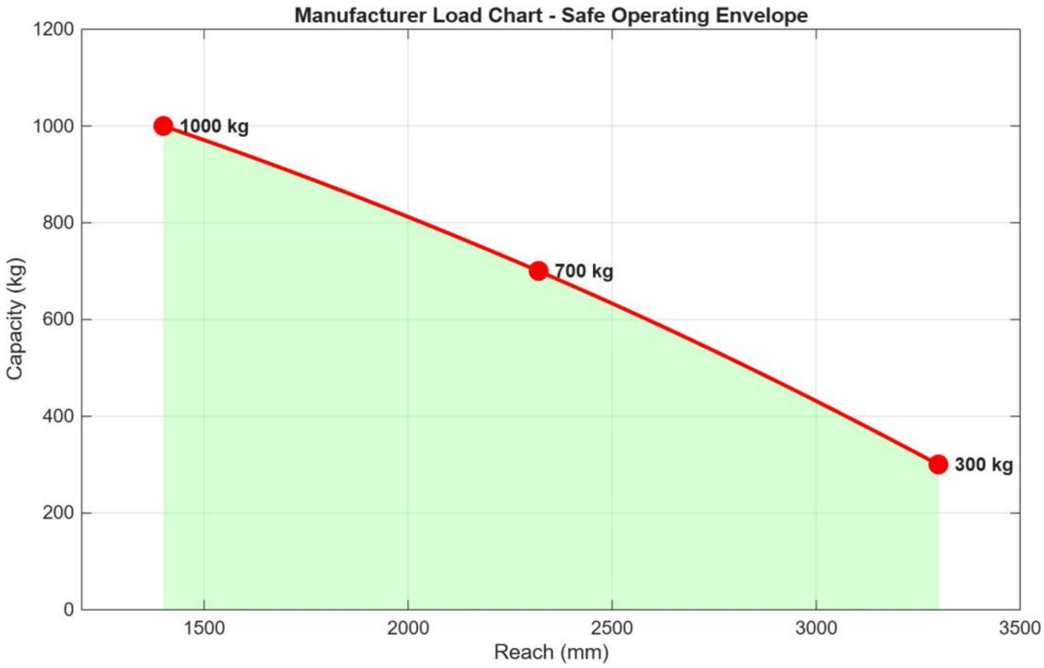


Figure 5.1 Rated load chart with the safe operating envelope shaded.

Chart point	Reach r (mm)	Rated load (kg)	Static load (kN)	Factored load W (kN)
Close	1400	1000	9.81	12.26
Mid	2320	700	6.87	8.58
Far	3300	300	2.94	3.68

5.2 Dynamic factor

A dynamic amplification factor of 1.25 is applied to the static load, consistent with HC1 hoist duty under EN 13001 for normal controlled lifting. The factored load is W = 1.25 x m x g. For the mid-reach point: W = 1.25 x 700 x 9.81 = 8.58 kN, as tabulated.

5.3 Which case governs

The design moment at the column is the factored load times its reach. Running the three points shows the mid-reach case is the most severe, even though it is not the heaviest load:

Design moment at column, M = W x r

```
Close   1000 kg:  M = 12.26 kN x 1.400 m = 17.17 kN.m
Mid      700 kg:  M =  8.58 kN x 2.320 m = 19.91 kN.m   <-- governs
Far      300 kg:  M =  3.68 kN x 3.300 m = 12.14 kN.m
```

Governing design moment: 19.91 kN m at 700 kg / 2320 mm.

This single result drives most of the report. The column, the base weld, the boom outer section and the stability check are all worst at this point. It is the case to keep in mind when reading the rest of the document.

6. Free Body and Design Forces

The free body for the governing case is shown below. The factored tip load reacts through the boom into the luffing cylinder and the pivot, and from there down the column to the base and the wheels. The counterweight and crane self-weight act at the rear.

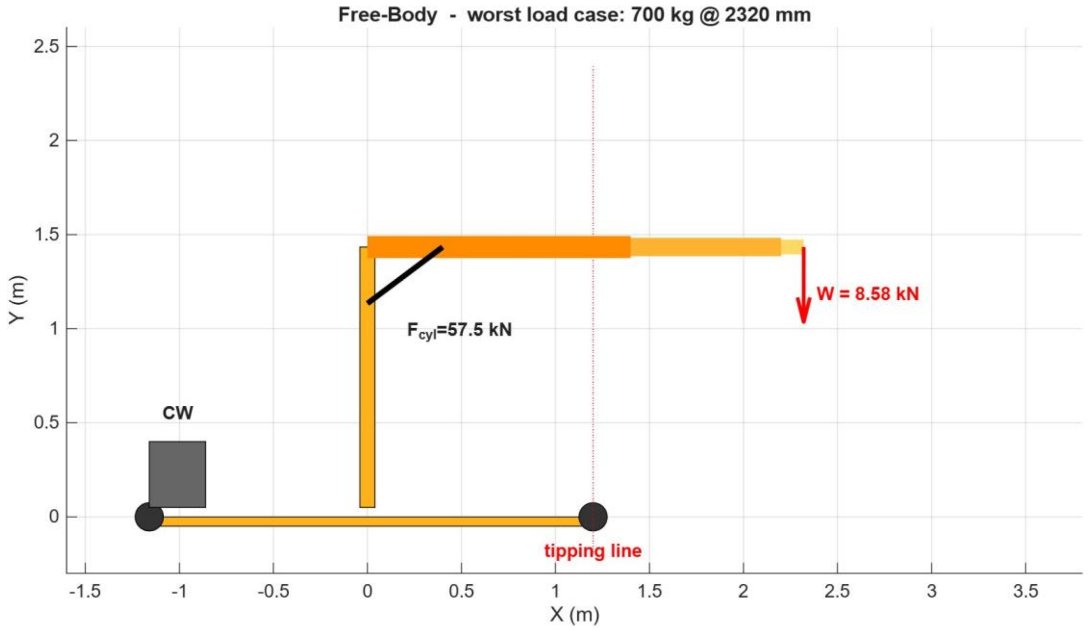


Figure 6.1 Free body at the governing case, 700 kg at 2320 mm. Tipping line at the front wheel, 1200 mm ahead of the column.

6.1 Cylinder and pivot forces

Taking moments about the boom pivot gives the cylinder force needed to hold the boom. The cylinder acts at a short lever arm relative to the load, so its force is large. The pivot then carries the vector sum of the cylinder force and the load reaction.

Boom moment balance about the pivot
Load about pivot: M_W = W x (lever to pivot)
Cylinder force: F_cyl = M_W / d_cyl = 57.49 kN

Pivot reaction (vector sum of F_cyl and W):
F_pivot = 63.85 kN

6.2 Wheel reactions

Resolving the whole crane about the wheel lines gives the support reactions. Almost all of the load goes to the front wheel pair, with the rear pair barely loaded at the governing case because the overturning moment has nearly lifted the rear off the ground.

Vertical and moment balance on the base
R_front (pair) = 15.16 kN -> 7.58 kN per wheel
R_rear (pair) = 0.29 kN -> 0.15 kN per wheel

Near-zero rear reaction confirms the crane is close to tipping
at this point, which the stability check (Section 7.6) quantifies.

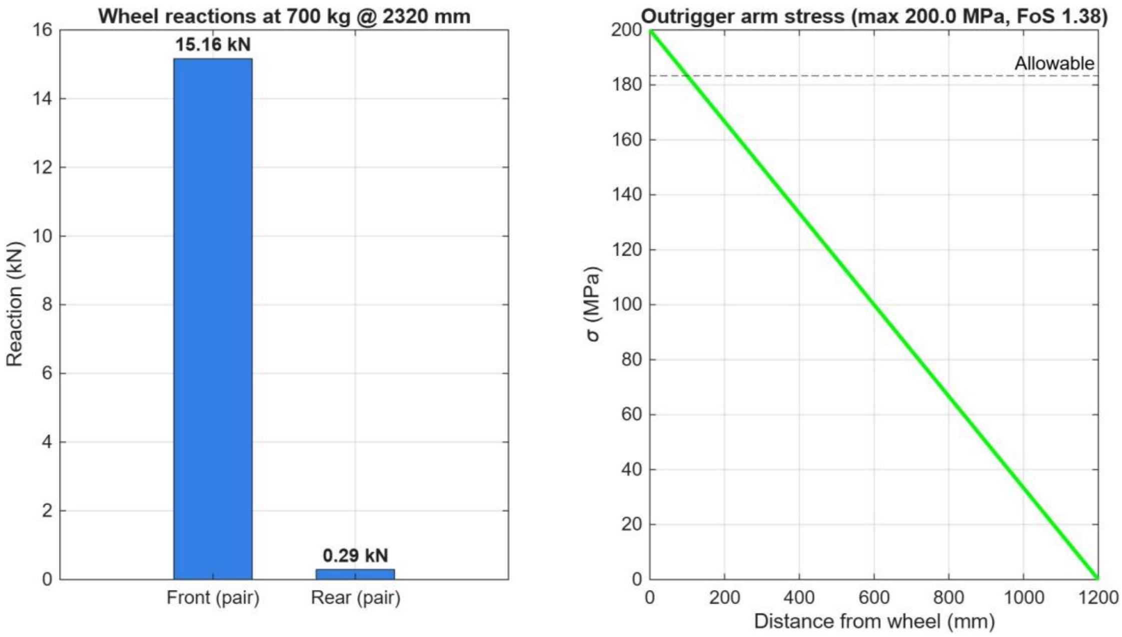


Figure 6.2 Wheel reactions and outrigger-arm stress profile at the governing case.

7. Detailed Calculations

Each check below shows the formula, the numbers put into it, the result, and the verdict against target. Stresses are compared to the 183 MPa working allowable for members, and pins and bolts to their 3.0 target.

7.1 Boom bending

The telescopic boom steps down in three sections from root to tip. Each section is checked at its own root, where its local moment is highest. The outer section at the column end carries the full design moment.

Boom outer (RSH 100 x 140 x 6) at root

M = 19.91 kN.m (governing)
Sx = 107.0 cm^3 = 107.0e3 mm^3
sigma = M / Sx = 19.91e6 / 107.0e3 = 186.2 MPa
FoS = fy / sigma = 275 / 186.2 = 1.48

FoS = 1.48 vs target 1.50 -> Marginal (just below target).

Boom middle (RSH 80 x 120 x 6) at its root

M = 7.90 kN.m
Sx = 73.0 cm^3
sigma = 7.90e6 / 73.0e3 = 108.1 MPa
FoS = 275 / 108.1 = 2.54

FoS = 2.54 -> Pass.

Boom inner (RSH 60 x 100 x 6) at its root

M = 4.05 kN.m
Sx = 45.5 cm^3
sigma = 4.05e6 / 45.5e3 = 89.0 MPa
FoS = 275 / 89.0 = 3.09

FoS = 3.09 -> Pass.

Tip deflection at the worst case reaches about 23 mm by hand against an L/200 serviceability guide of roughly 12 mm, so the boom is also outside the deflection guide at full reach. The frame model gives a larger tip movement because it includes column and base flexibility; this is discussed in Section 10.

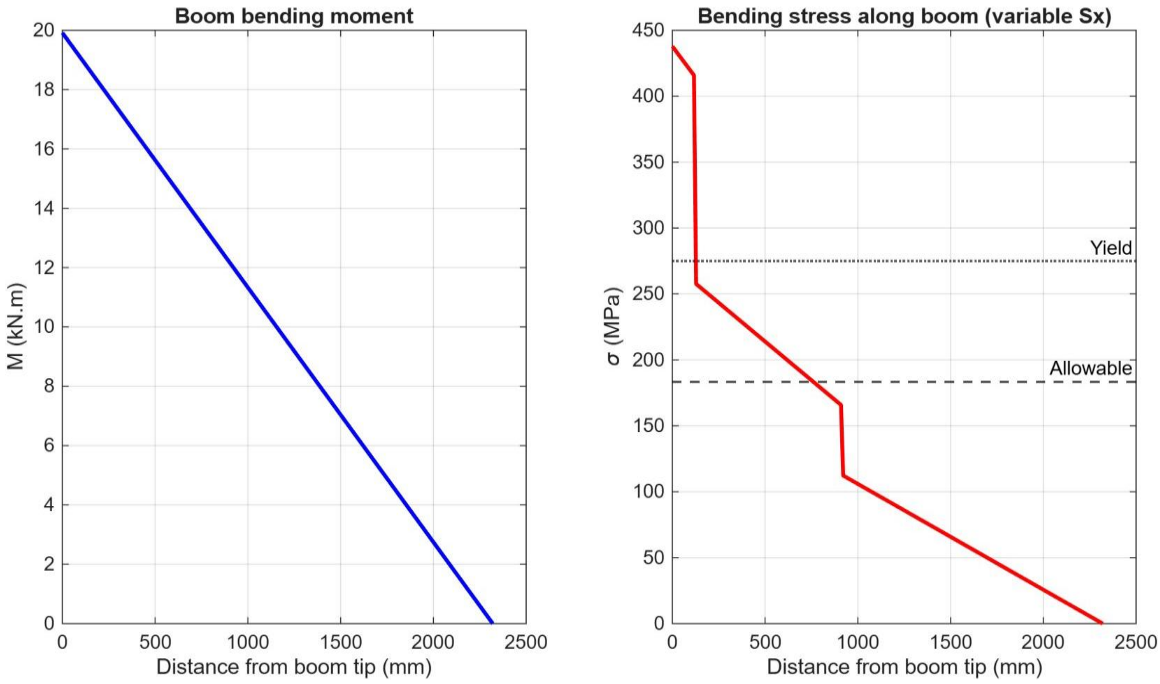


Figure 7.1 Boom bending moment (left) and stress along the boom with the stepped section (right). The outer section sits above the 183 MPa allowable.

7.2 Column under combined load

The column carries axial load from the boom system plus the full bending moment transferred at its top. The two stresses add directly. This is the governing structural check.

Column (RSH 80 x 120 x 6), combined axial + bending

N = 10.64 kN A = 22.6 cm^2 = 2260 mm^2
M = 19.91 kN.m Sx = 73.0 cm^3 = 73.0e3 mm^3

sigma_axial = N / A = 10.64e3 / 2260 = 4.7 MPa
sigma_bend = M / Sx = 19.91e6 / 73.0e3 = 272.7 MPa
sigma_total = 4.7 + 272.7 = 277.4 MPa

FoS = fy / sigma_total = 275 / 277.4 = 0.99

FoS = 0.99 vs target 1.50 -> Fail. Stress is at yield.

Column buckling (Euler, weak axis, K = 2 cantilever)

Slenderness lambda = 80.6
Critical load Pcr = 720 kN
Applied axial N = 10.64 kN
FoS_buckling = 720 / 10.64 = 67.6

FoS = 67.6 -> Pass. Buckling is not a concern; bending governs.

Reading the column result

The column is a bending problem, not a buckling problem. Its axial stress is tiny (under 5 MPa) and buckling has a vast margin, but the bending stress alone uses up the entire yield capacity. To recover a 1.5 margin the section modulus would need to rise by about half, which points to a deeper or thicker column section rather than any change to axial capacity.

7.3 Base frame and outrigger arm

The outrigger arm is checked as a cantilever picking up the front wheel reaction at its tip. The moment at the root gives the stress.

Outrigger arm (RSH 60 x 100 x 6)

M_arm = 9.10 kN.m Sx = 45.5 cm^3
sigma = 9.10e6 / 45.5e3 = 200.0 MPa
FoS = 275 / 200.0 = 1.38

FoS = 1.38 vs target 1.50 -> Fail (above the 183 MPa allowable).

7.4 Pins and connections

The pins are checked in shear (double or single as built) and in bearing on the lug. The cylinder force and pivot reaction from Section 6.1 drive the pin loads.

Boom pivot pin, dia 30 mm, double shear

F_pivot = 63.85 kN shear area = 2 x pi/4 x 30^2 = 1414 mm^2
tau = 63.85e3 / 1414 = 45.2 MPa
bearing = 141.9 MPa
FoS (as-coded, against 110 MPa basis) = 2.44

FoS = 2.44 vs target 3.00 -> Fail on the conservative basis. See note below.

Cylinder rod-end pin, dia 20 mm, double shear

F_cyl = 57.49 kN shear area = 2 x pi/4 x 20^2 = 628 mm^2
tau = 57.49e3 / 628 = 91.5 MPa
bearing = 191.6 MPa
FoS = 1.20

FoS = 1.20 vs target 3.00 -> Fail on any basis. The genuine weak pin.

Hook pin, dia 20 mm, single shear
tau = 27.3 MPa bearing = 28.6 MPa
FoS = 4.03
FoS = 4.03 -> Pass.

Basis note for the reviewer (pins)
The script applies the material factor (yield/1.5) and the 3.0 pin factor in series, which is conservative. On the conventional lifting-pin basis where shear capacity is 0.6 x fy and a 3.0 factor is required against that, the pivot pin moves from 2.44 to about 3.65 and passes, and the hook pin from 4.03 to about 6.04. The cylinder pin still fails, moving from 1.20 to about 1.80. The summary table keeps the conservative as-computed figures so nothing contradicts the issued numbers, but the reviewer should decide which pin basis applies. The clear and basis-independent finding is that the cylinder rod-end pin is undersized.

7.5 Welds

Fillet welds are checked on the throat area at 6 mm leg. The column-base weld carries the full moment as a stress couple across the weld group and is the governing weld.

Column-to-base weld (6 mm fillet)
sigma_weld = 234.9 MPa
FoS = 0.85
FoS = 0.85 -> Fail. Governs alongside the column itself.

Boom-bracket weld (6 mm fillet)
tau_weld = 35.5 MPa
FoS = 3.39
FoS = 3.39 -> Pass.

Outrigger weld (6 mm fillet)
sigma_weld = 170.3 MPa
FoS = 1.18
FoS = 1.18 -> Fail.

7.6 Slewing-flange bolts

The five M16 grade 8.8 bolts sit on a 170 mm pitch circle. The overturning moment is shared as a tension couple across the pattern using the polar-moment method, and combined tension-plus-shear is checked against the EN 1993-1-8 interaction.

5 x M16 grade 8.8 on 170 mm PCD, polar-moment method
Bolt capacity: tension 113.0 kN, shear 75.4 kN
Peak demand: tension 93.7 kN, shear 2.1 kN per bolt
Interaction utilisation = 0.35
FoS (combined) = 1.69
FoS = 1.69 vs target 1.50 -> Pass.

7.7 Stability against overturning

Stability is a moment balance about the front wheel line, 1200 mm ahead of the column. The restoring moment comes from the crane self-weight and counterweight acting behind that line; the overturning moment comes from the load acting ahead of it. The mid-reach point governs.

Overturning balance about the front wheel line
Restoring M_R = m_crane x g x (L_fwd + e_cg)
= 700 x 9.81 x (1.200 + 0.300) = 10.30 kN.m

```
Load case           M_OT (kN.m)   FoS = M_R/M_OT
  1000 kg @ 1400 mm     1.96        5.25   Pass
   700 kg @ 2320 mm     7.69        1.34   Fail  <-- governs
   300 kg @ 3300 mm     6.18        1.67   Pass
```

Worst stability FoS = 1.34 at 700 kg / 2320 mm vs target 1.50 -> Fail (wheels only).

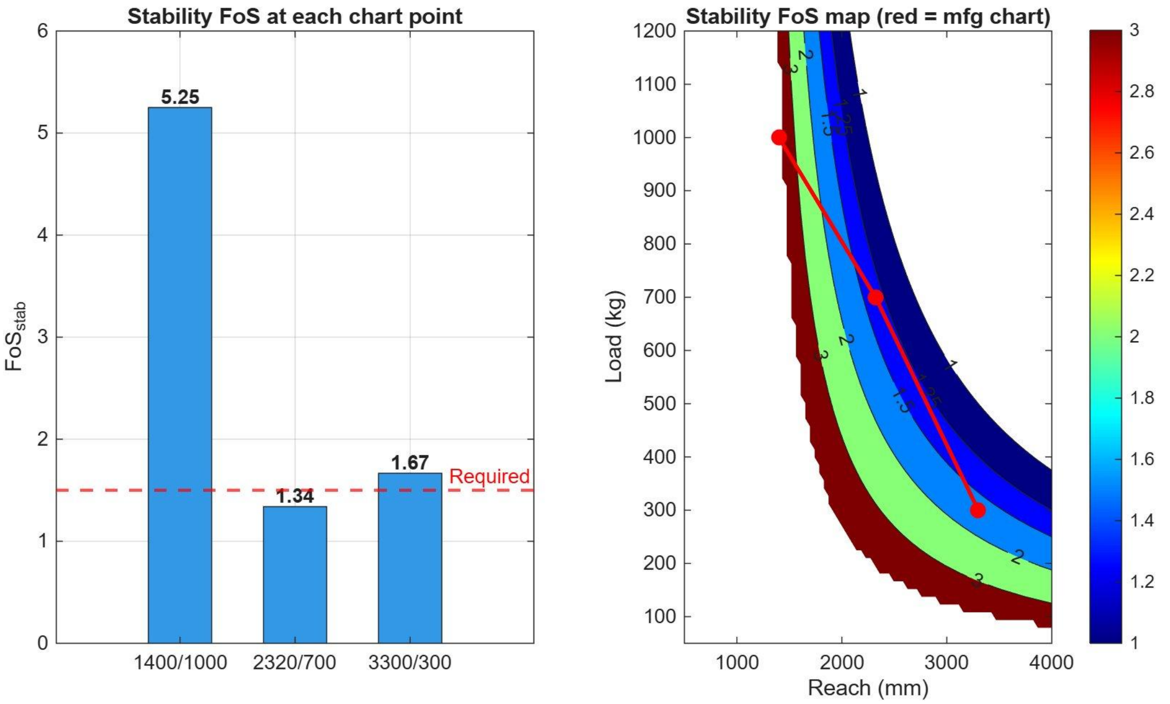


Figure 7.2 Stability factor at each chart point (left) and the factor map across reach and load with the chart overlaid (right).

8. Frame Model Results

The plane-frame solver was run for all three chart points. The deformed shape and the stress distribution for the governing case are shown below. The model confirms the column as the most-stressed member, in agreement with the hand calculation.

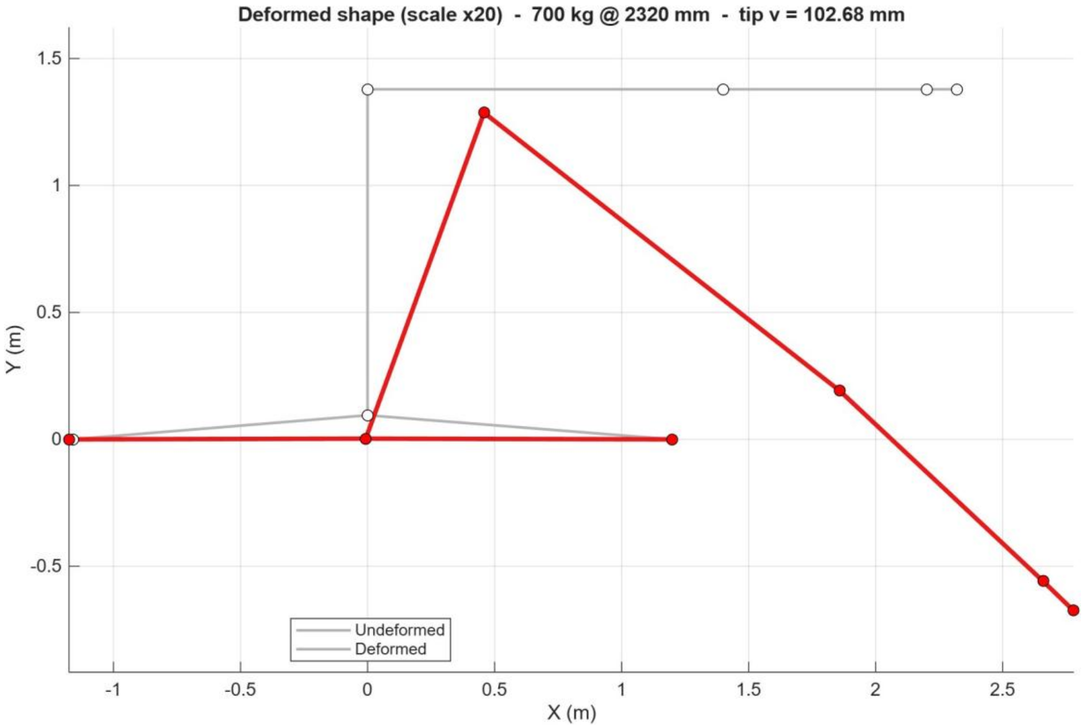


Figure 8.1 Deformed shape at the governing case, scaled x20. Boom tip movement reflects column and base flexibility, not boom bending alone.

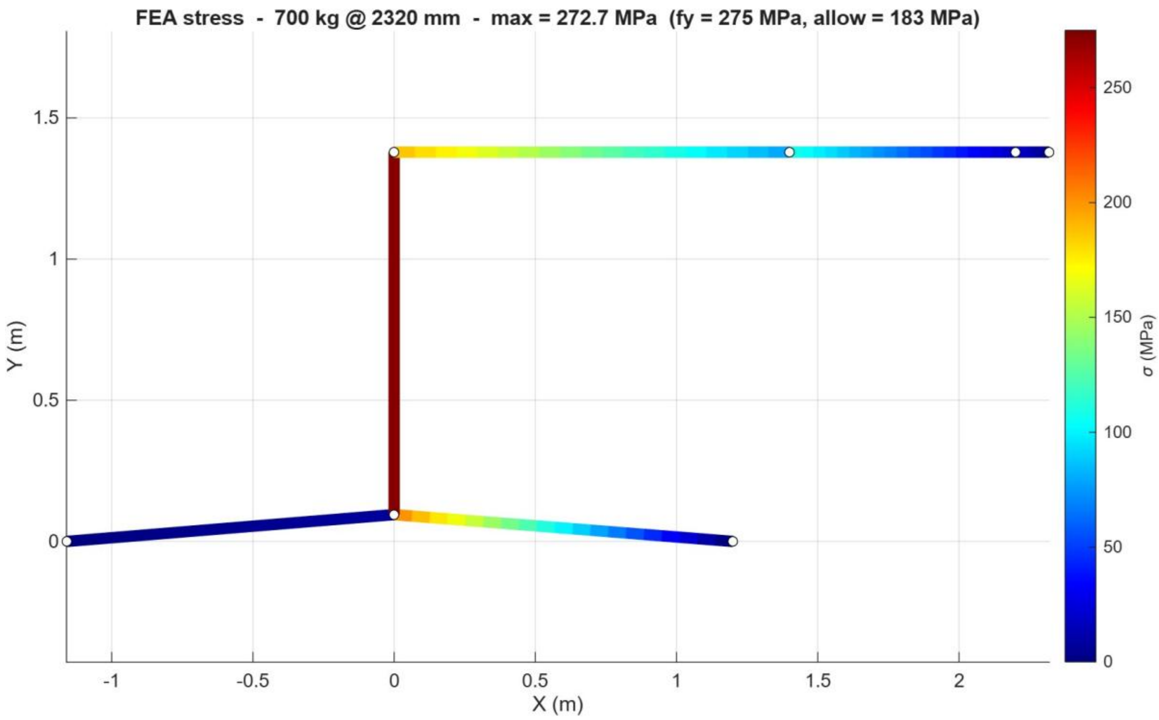


Figure 8.2 Frame stress at the governing case. Peak 272.7 MPa at the column, against yield 275 MPa and the 183 MPa allowable.

Chart point	Tip deflection (mm)	Peak stress (MPa)	FoS

1000 kg @ 1400 mm	51.5	235.1	1.17
700 kg @ 2320 mm	102.7	272.7	1.01
300 kg @ 3300 mm	101.1	166.2	1.65

The frame FoS values are quoted against yield directly (275 / peak stress), which is why they read slightly higher than the 1.5-target factors in the hand calc. On a like-for-like basis the two methods agree, as Section 10 shows.

9. Operating Envelope

The script also sweeps the boom through its luffing range and tracks the boom-root stress and factor of safety against the chart capacity at each angle. The still below is taken from that sweep. It shows the safe capacity dropping at the reaches where stress approaches the allowable line.

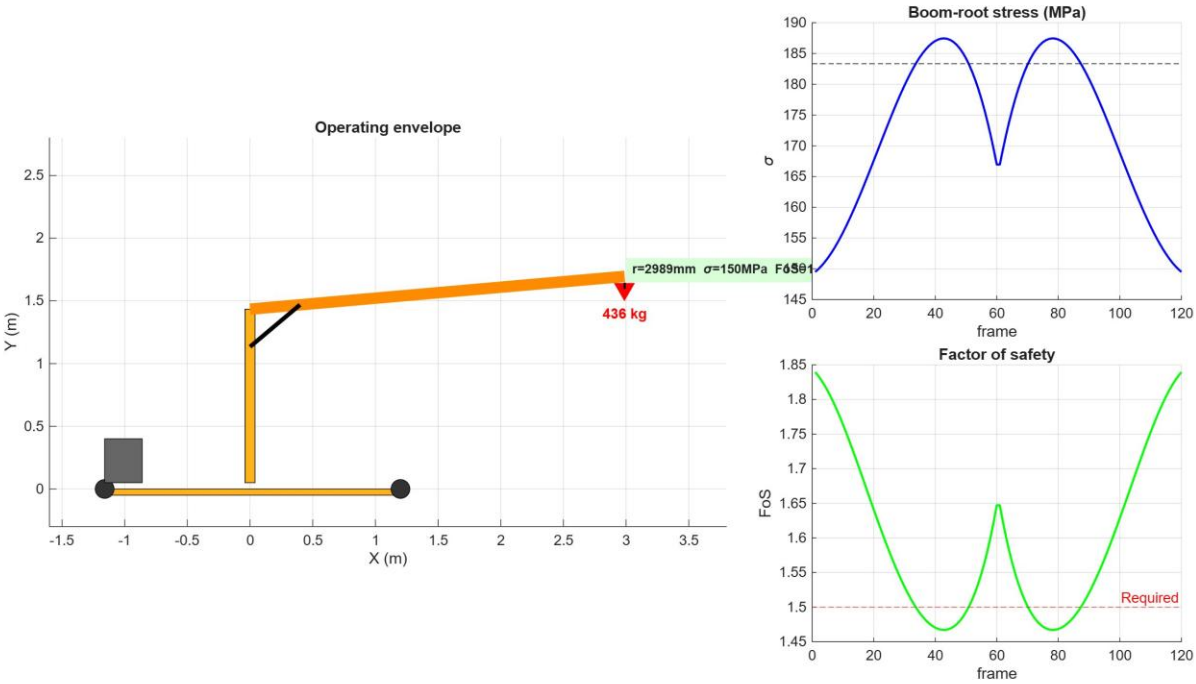


Figure 9.1 Operating-envelope sweep. Left: crane pose. Right: boom-root stress and factor of safety through the luffing cycle, with the 183 MPa allowable and 1.5 target marked.

10. Cross-Validation of the Two Methods

The hand calculation and the frame model are compared on the three quantities they both produce. The column-base moment matches exactly, and the peak stress agrees to within two percent. The tip deflection differs because the two are measuring different things.

Quantity	Track 1 (hand)	Track 2 (frame)	Agreement
Column-base moment (kN m)	19.91	19.91	Exact
Peak member stress (MPa)	277.4	272.7	Within 2%
Boom tip deflection (mm)	22.7	102.7	Different basis

The deflection gap is expected and not an error. The hand figure is boom bending alone, treating the column top as rigid. The frame figure adds the column bending and the base flexibility, so the tip moves much further. For a stress verdict the moment and stress agreement is what matters, and that is tight.

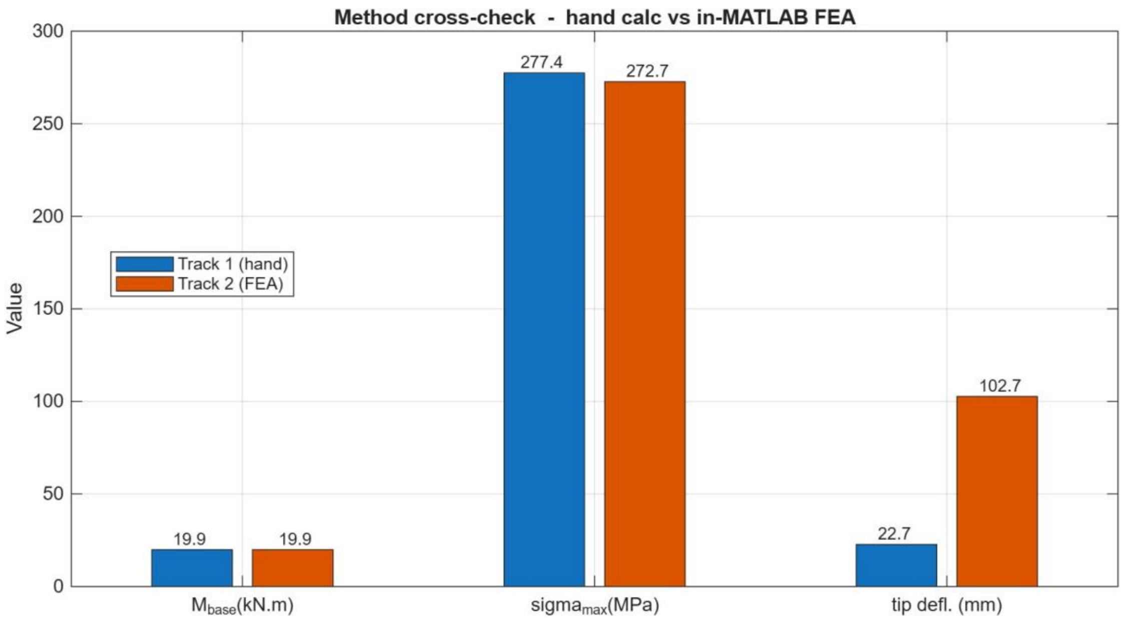


Figure 10.1 Side-by-side comparison of the two methods.

11. Summary of Results

All fourteen checks are collected below with their factors of safety. Failing items are shaded red, passing items green, and the boom outer section amber as a borderline miss. Eight of fourteen do not meet target.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.48	1.50	**Marginal**
Boom middle (80x120x6)	2.54	1.50	**Pass**
Boom inner (60x100x6)	3.09	1.50	**Pass**
Column, combined N+M	0.99	1.50	**Fail**
Column buckling	67.62	1.50	**Pass**
Outrigger arm	1.38	1.50	**Fail**
Pivot pin	2.44	3.00	**Fail**
Cylinder pin	1.20	3.00	**Fail**
Hook pin	4.03	3.00	**Pass**
Slewing bolts	1.69	1.50	**Pass**
Weld, column base	0.85	1.50	**Fail**
Weld, boom bracket	3.39	1.50	**Pass**
Weld, outrigger	1.18	1.50	**Fail**
Stability (worst point)	1.34	1.50	**Fail**

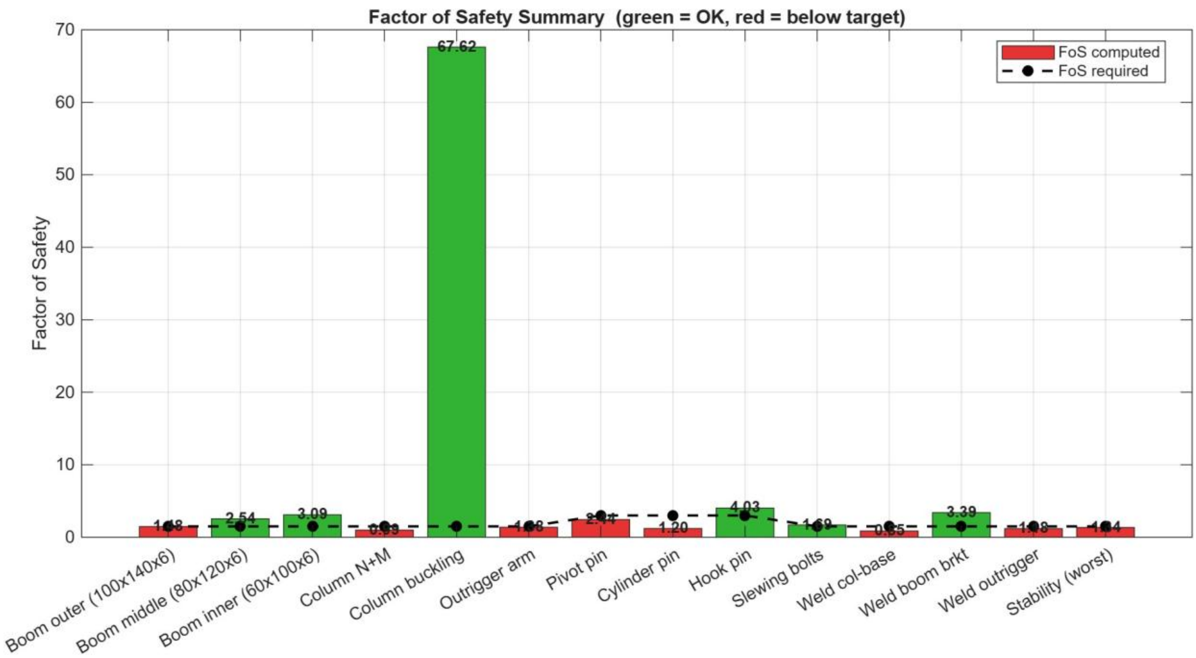


Figure 11.1 Factor of safety for every check against the 1.5 / 3.0 targets.

12. Conclusions and Recommendations

12.1 Conclusion on the 1 tonne SWL

In the wheels-only configuration the crane does not meet the agreed factors of safety across its rated chart. At the close-in 1400 mm position the steel carries the 1 tonne load below yield, but only to a factor of 1.17 against the 1.50 target. At the governing 700 kg mid-reach point the column reaches yield (FoS 0.99) and its base weld is past target (FoS 0.85). The crane therefore cannot be certified to the chart as drawn.

Put plainly for the operator: the crane will lift the loads, but it does not carry the safety margin a rated 1 tonne crane is expected to have, and at medium reach it is at the limit of the steel. The lighter mid-reach lift is the dangerous one, not the heavy close lift, because of the longer lever arm.

12.2 The governing items

Two items genuinely govern and two more are borderline:

- **Column, FoS 0.99.** At yield in bending. The clear primary failure.
- **Column-base weld, FoS 0.85.** Past target on the same load path. Fails with the column.
- **Cylinder rod-end pin, FoS 1.20.** Undersized on any basis, independent of the others.
- **Boom outer and outrigger arm, FoS 1.48 and 1.38.** Marginal misses that the recommended actions resolve as a by-product.

12.3 Three ways forward

Route A: re-rate the chart (recommended primary action)

Drop the rated capacities so every check clears 1.5. Back-calculation puts the safe envelope near 400 to 450 kg at the mid and far reaches and around 800 kg in close. This is the cheapest route, needs no fabrication, and is the honest reflection of what the structure as built safely carries.

Route B: ease the dynamic factor

If the duty is genuinely slow and controlled indoor lifting, EN 13001 allows a lower dynamic factor near 1.10. That lifts the marginal items but does not on its own rescue the column or the base weld, so it is a supporting measure rather than a fix.

Route C: strengthen the column and base (keeps full rating)

To hold the 1 tonne chart the column needs a larger section, on the order of 100 x 150 x 8 in S275, which raises its modulus enough to bring the bending FoS back above 1.5. The base weld must grow to suit, and the cylinder rod-end pin should go up one size. This is the largest job but the only route that preserves capacity.

Recommendation
Adopt Route A as the immediate action so the crane can be used safely at a reduced, verified rating, and hold Route C as the upgrade path if the full 1 tonne chart must be retained. Whichever route is chosen, the crane should not be worked to the current chart until the column, base weld and cylinder pin are addressed.

13. Assumptions and Items to Confirm

The section sizes, lengths, material grade, weld throats, bolt grade and slewing pattern are taken directly from the drawings and are not in doubt. The following inputs are estimates or client-supplied values and should be confirmed against the built machine before the report is relied on for certification.

- **Crane self-weight, 700 kg.** Client-supplied with no bill of materials. A swing of plus or minus 100 kg moves the stability result either way.
- **Centre-of-gravity offset, 0.30 m behind the column.** Estimated from the counterweight and pump positions. Affects stability only; does not touch the governing column or weld checks.
- **Front overhang to tipping line, 1200 mm.** Derived from the base drawing dimension chain. Used for stability and wheel reactions.
- **Weld throats, 6 mm.** Taken as the drawing minimum. If site welds are smaller the weld checks worsen.
- **Pin verification basis.** See Section 7.4. The reviewer should confirm whether the conservative series-factor basis or the conventional lifting-pin basis applies to the pin verdicts.

14. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion Files

- MATLAB R2026a parametric script (jib_crane_verification.m), single self-contained file driving every number in this report
- Console log (console_log.txt), full text of the run including the factor-of-safety table
- Figures 1 to 10 as PNG, and the operating-envelope animation as MP4
- workspace.mat, saved variables for re-plotting without re-running

Appendix B: Result Index by Figure

Figure	Content
Fig 6.1 / title	Free body at governing case
Fig 7.1	Boom bending moment and stress
Fig 8.1	Frame deformed shape
Fig 8.2	Frame stress distribution
Fig 5.1	Rated load chart
Fig 6.2	Wheel reactions and outrigger stress
Fig 7.2	Stability factor and map
Fig 11.1	Factor of safety summary
Fig 10.1	Two-method cross-check
Fig 9.1	Operating-envelope sweep